

Capstone Introduction and Understanding the Datasets

Project scenario and Overview

The commercial space age is here, companies are making space travel affordable for everyone.

- Virgin Galactic is providing suborbital spaceflights.
- Rocket Lab is a small satellite provider.
- Blue Origin manufactures sub-orbital and orbital reusable rockets.
- SpaceX's accomplishments include:

Sending spacecraft to the International Space Station.

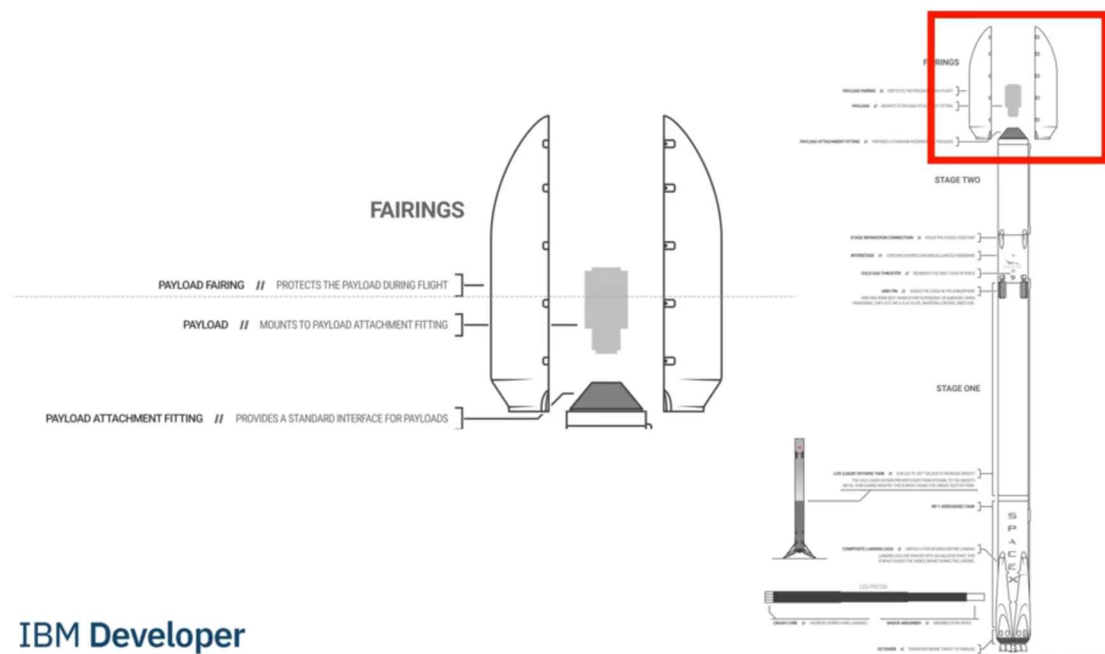
Starlink, a satellite internet constellation providing satellite Internet access.

Sending manned missions to Space.

One reason SpaceX can do this is the **rocket launches are relatively inexpensive**.

SpaceX advertises Falcon 9 rocket launches on its website with a cost of 62 million dollars; other providers cost upwards of 165 million dollars each, much of the **savings is because SpaceX can reuse the first stage**. Therefore, if we can determine if the first stage will land, we can determine the cost of a launch.

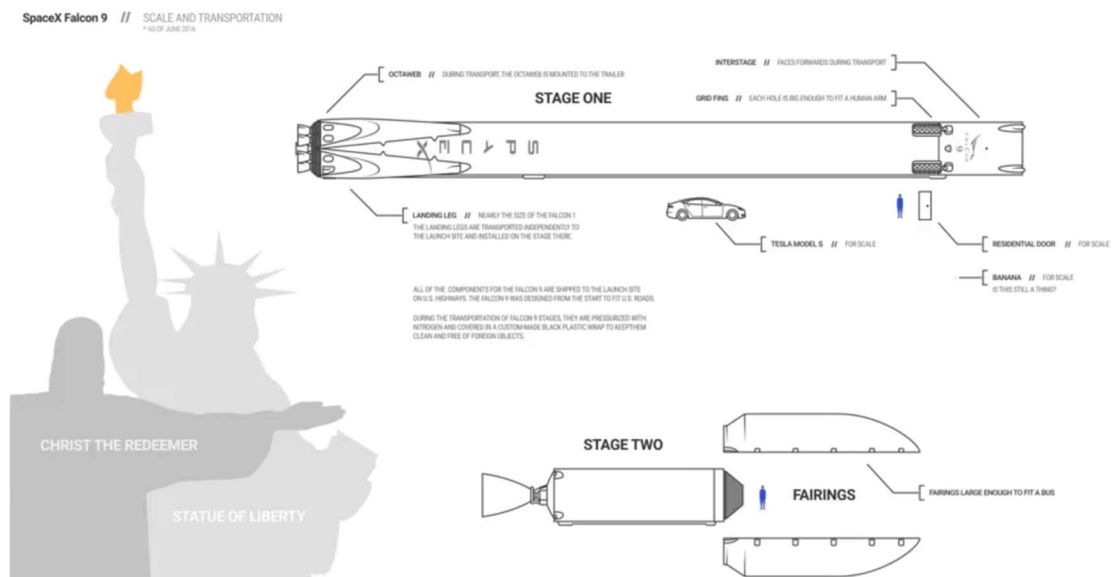
SpaceX's Falcon 9 launch like regular rockets. To help us understand the scale of the Falcon 9, we are going to use these diagrams from Forest Katsch, at zlsdesign.com. He is a 3D artist and software engineer. He makes infographics on spaceflight and spacecraft art. He also makes software.



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The payload is enclosed in the fairings. Stage two, or the second stage, helps bring

the payload to orbit, but most of the work is done by the first stage. The first stage is shown here. This stage does most of the work and is much larger than the second stage.



Here we see the first stage next to a person and several other landmarks. This stage is quite large and expensive. Unlike other rocket providers, SpaceX's Falcon 9 Can recover the first stage. **Sometimes the first stage does not land. Sometimes it will crash** as shown in this clip. Other times, Space X will sacrifice the first stage due to the mission parameters like payload, orbit, and customer.

In this capstone, **you will take the role of a data scientist** working for a new rocket company. Space Y that would like to compete with SpaceX founded by Billionaire industrialist Allon Musk.

Your job is to **determine the price of each launch**. You will do this by gathering information about Space X and creating dashboards for your team. You will also determine if SpaceX will reuse the first stage.

Instead of using rocket science to determine if the first stage will land successfully, you will train a machine learning model and use public information to predict if SpaceX will reuse the first stage.